NUS SDS X DSESC *MINI-DATATHON* 2026

BRIEFING

CASE BRIEF

POLICY CONTEXT

- Singapore controls its car population through the Certificate of Entitlement (COE) system, introduced in 1990[1]

- Since 1999, cars have been split into 2 COE categories[1], by engine size, and since 2014, by engine power[2]
  - Cat A for mass-market cars
  - Cat B for higher-value cars
- The split is to stop luxury buyers from pricing out ordinary buyers, BUT electric vehicles can be tuned to meet Cat A's power limit[3]

[1] Dollars and Sense. History of the COE, and how it has evolved since 1990. dollarsandsense.sg
[2] Land Transport Authority. Annual Motor Vehicle Population by Vehicle Quota Categories. data.gov.sg
[3] Land Transport Authority. (2026, October 8). Public consultation on COE categorisation [Press release]. lta.gov.sg

CASE BRIEF

POLICY CONTEXT

In response, LTA opened its proposal to merge Categories A and B into one car category, up for public consultation[4]

- "Feebate" is based on each car model's median Open Market Value (OMV): cheaper cars get a rebate on their COE and pricier cars pay a surcharge[1].
- LTA's examples use rebates and surcharges of up to 15,000[5], in either a three-band or a five-band structure[4]
- LTA is also asking what should happen to Category E and whether the fee-rebate should apply to COE renewals[4]

[4] Land Transport Authority. (2026, October 8). Public consultation on COE categorisation, Annex A: At a glance [Press release]. lta.gov.sg
[5] Land Transport Authority. (2026, October 8). Public consultation on COE categorisation [Consultation paper]. lta.gov.sg

CASE BRIEF

PROBLEM STATEMENT

You are an independent data science team **advising LTA as it decides the COE system's future**. Assess whether the current Category A/B split **still separates mass-market cars from higher-value cars**, and **recommend** whether LTA should **merge the categories, keep them, or adopt another design.**

DATASET 1, Land Transport Authority (Jan 2010 – Oct 2026)

COE BIDDING RESULTS / PRICES

Month Month (YYYY-MM)		Bidding No Text		Vehicle Class Text		Quota Text		Bids Success Text		Bids Received Text		Premium Text	
(Null)	0.0%	(Null)	0.0%	(Null)	0.0%	(Null)	0.0%	(Null)	0.0%	(Null)	0.0%	(Null)	0.0%
2010-01	0.5%	1	50.1%	Category A	20.0%								
2010-02	0.5%	2	49.9%	Category B	20.0%								
2010-03	0.5%			Category C	20.0%								
2010-04	0.5%			Category D	20.0%								
195 more values				1 more value									
2010-02		1		Category A		1154		1153		1326		19989	
2010-02		1		Category B		1154		690		930		23180	

* Results of Singapore's COE bidding exercises: quotas, bid volumes, successful bids, and clearing premiums across vehicle classes/categories A–E.

Dataset 1 is the main dataset for analysing the Cat A–B premium gap, bidding pressure, forecasting future premiums and simulating a merged-category system.

DATASET 2, Land Transport Authority (Jan 2016 - May 2025)

NEW REGISTRATION OF CARS BY MAKE

Month Month (YYYY-MM)		Make Text		Importer Type Text		Fuel Type Text		Vehicle Type Text		Number Numeric	
(Null)	0.0%	(Null)	0.0%	(Null)	60.3%	(Null)	0.0%	(Null)	0.0%	(Null)	57.0%
2023-06	1.7%	MERCEDES BENZ	5.4%	AMD	26.8%	Petrol	52.0%	Sedan	18.5%	0	12.4%
2023-05	1.6%	B.M.W.	4.7%	PI	12.9%	Electric	11.8%	Hatchback	18.5%	1	6.2%
2023-01	1.6%	TOYOTA	4.5%			Diesel	11.4%	Coupe/ Convertible	18.5%	2	3.0%
2023-02	1.6%	AUDI	3.4%			Others	11.2%	Sports Utility Vehicle	18.5%	3	1.9%
109 more values		82 more values				4 more values		3 more values		481 more values	
2016-01		ALFA ROMEO				Petrol		Hatchback		3	
2016-01		ALPINA				Petrol		Hatchback		0	

* Measuring the number of newly registered cars in Singapore, broken down by manufacturer, fuel type, importer type and vehicle body type.
** Does not directly identify the COE category of each registered car.

Dataset 2 helps identify which brands and fuel types are gaining market share, particularly EVs, and whether these shifts may be associated with demand for Category A or B COEs.

CASE BRIEF

EXPECTED OUTCOMES



Trend analysis: Examine whether the gap between Category A and B COE premiums has narrowed over time, identifying key trends in premiums and bidding demand to determine whether recent convergence reflects a lasting shift or a temporary fluctuation.



Forecasting: Develop models to predict Category A and B premiums for upcoming bidding exercises, evaluating their accuracy against simple baseline methods such as previous-round premiums or rolling averages.



Market Insights: Explore new vehicle registrations, and identify emerging trends in fuel types and car brands, assessing how changes in market share relate to bidding demand and competition within each COE category.

CASE BRIEF

EXPECTED OUTCOMES

Merger Feasibility: Evaluate whether the evidence supports merging Categories A and B, comparing historical premiums with a hypothetical merged premium and projecting future outcomes under clearly defined assumptions.

Policy Recommendations: Propose data-driven recommendations on the feasibility and implications of a category merger, considering its potential impact on different groups of car buyers.

Bonus: Participants supporting a feebate system are encouraged to compare three-band and five-band designs, evaluating their effectiveness in supporting mass-market buyers and the trade-offs involved.

CASE BRIEF

MODELLING GUIDELINES

Target

- Public data shows round outcomes, not individual bids
- Consider modelling the clearing premium or a measure derived from it
  - e.g. Cat A-B premium gap, probability Cat A exceeds Cat B, premium volatility

Suggested Features

- COE data: quota, bids received, bids-to-quota ratio, previous premiums, bidding round
- Registration data: EV share, fuel mix, brand mix
- Aggregate registrations to monthly totals, join on Month
- Bids received are only known after a round closes
  - Use to explain premiums rather than forecast premiums

CASE BRIEF

MODELLING GUIDELINES

Validation

- Use time-based splits
- Compare against simple baselines (previous round's premium, rolling average)
- Account for structural breaks

Additional Considerations

- Merged-category premiums are not observable
  - State your assumption (e.g. quota-weighted average of A and B) and test how sensitive your conclusions are to it

CASE BRIEF

JUDING CRITERIA

Recommendation & Relevance (20%)

- A clear, actionable answer that considers the policy's trade-offs and affected groups

Data Visualisation (25%)

- Well-chosen, clearly labelled charts that reflect the points argued

Analytical Rigour (25%)

- Use of sound methods, good data cleaning and reasoning

Transparency & Limitations (15%)

- Evidence, assumptions and speculation clearly separated; specific limitations stated

Communication (15%)

- Logical link between evidence to recommendation

CASE BRIEF

SUBMISSION FORMAT

Deadline: Friday, 30 October, 2:00 PM SGT

- Upload a .zip file named [TeamName].zip (max 200MB) via the Google Form link (on the next slide)
- Your zip file must contain:

TeamName/
- **main.ipynb**
- **main.html**
- **requirements.txt** (List of dependencies/libraries used)
- **deck.pdf** (Slide deck)
- **data/**

- Slide deck (max 12 slides):
  - Stating your recommendation clearly up front
  - Evidence, with charts generated from the data
  - Separating what the data shows, what you assumed, and what is speculation
  - Limitations of your analysis

CASE BRIEF

SUBMISSION FORM



https://tinyurl.com/yvhcep6y

CASE BRIEF

FAQ

Which dataset is more important?
The COE bidding data is the core dataset, since it drives the premium you're modelling. The registration data helps explain premium trends, e.g., through EV share or fuel mix.

Can I use additional datasets?
Yes, but your main analysis must use the provided datasets. Additional data may only support your argument. It must come from official public sources (e.g. data.gov.sg, LTA, SingStat), be included in data/, and be cited with its source and download date.

Can I use COE data from before 2016?
Yes, for analysis that uses only the COE data. The Jan 2016 – May 2025 limit applies only when joining with the registration data.

CASE BRIEF

FAQ

Can I use R, Excel or Tableau?
No. All analysis and charts must be produced in Python in main.ipynb.

Can I resubmit before the deadline?
Yes, your latest submission before the deadline will be graded

THANK YOU!